The Link Function Problem in Psychological Interaction Testing

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Abstract

Equal differences on one scale become different differences on another scale. This is crucial for interaction testing because an interaction is a difference between differences, and in psychological research the observed response and the construct of interest are often expressed on different scales. Therefore, the same data can show an interaction on one scale and not on another. Generalized Linear Models make this problem explicit by separating two decisions: the family, which describes the nature of the response variable, and the link function, which sets the scale on which effects are additive. Because a model without an interaction term is additive on the link scale, choosing a link is choosing what 'no interaction' means. The paper has two connected parts. First, a preregistered review of recent articles from five prominent psychology journals shows that interaction testing is routinely performed in research, that the link function is most often unspecified, and that constrained outcomes are frequently analyzed with gaussian identity-link models. Second, worked examples and simulations develop three common response formats: forced-choice accuracy with a chance floor, within-family link choices (logit vs probit), and sum scores as a measurement case, and quantify how a plausible-looking link can turn an additive structure on one scale into a statistically significant interaction on another. A final diagnostic simulation shows that residual checks, a link test, and AIC comparisons give uneven protection and do not by themselves decide which scale the scientific claim concerns. Interaction claims remain underspecified until researchers state, justify, and probe the scale on which no interaction is assumed.

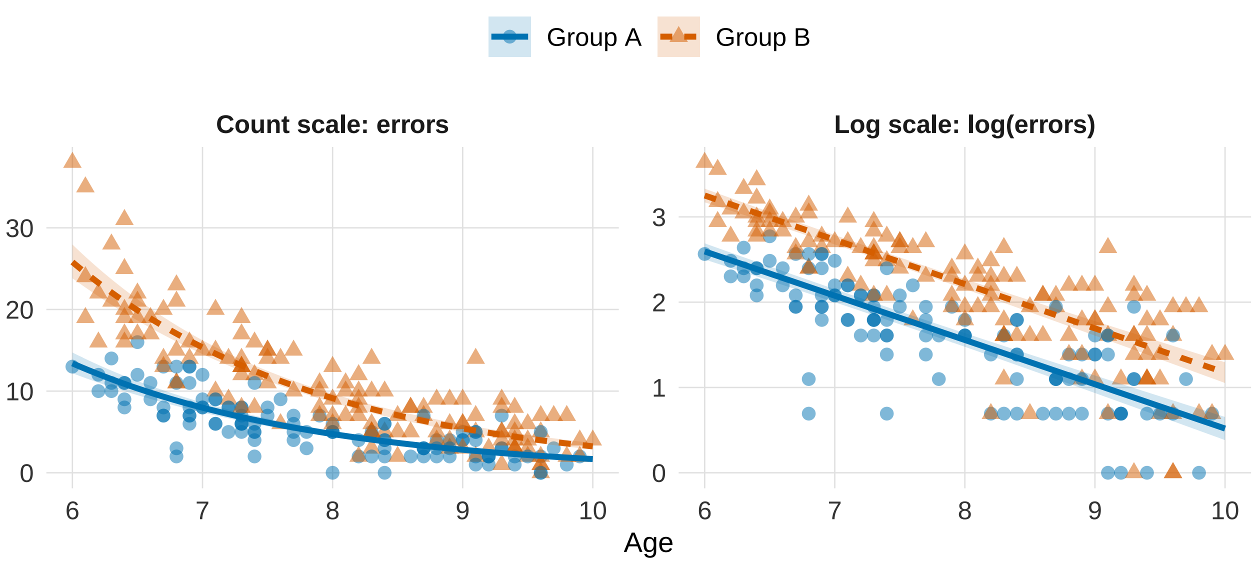
Keywords: interactions, moderation, generalized linear models, link functions, model misspecification

The Link Function Problem in Psychological Interaction Testing

We start with a very simplified example. Two populations of children (e.g., clinical vs neurotypical) differ in their overall error rates on a task, and errors decrease with age in both. Figure 1 samples simulated data from such a scenario. In the left panel (observed response), errors decline with age in both groups, and the difference between groups is much larger at age six than at age ten. Is there an age-by-group interaction here?

Figure 1

Simulated error counts for two groups of children aged 6 to 10, shown on the observed count scale (left) and on the log scale (right). The two groups have exactly the same age effect on the log of the expected error count, so on the log scale the two fitted lines are parallel, with a constant group gap at every age. Mapped back onto the count scale, the same model produces converging curves, so the group difference shrinks with age. Whether this pattern is an interaction depends on the scale on which effects are assumed to add: on the count scale the group difference changes with age; on the log scale it does not.



Because an interaction is a difference between differences, the question is whether the group difference itself changes with age. On the observed count scale, the group effect does appear to shrink as age increases, so it looks like an interaction. But looking at the same data on the log-scale (Figure 1, right panel), both groups have the same age slope, meaning their fitted lines are parallel. A constant difference on the log scale corresponds to a constant ratio on the count scale, so the absolute gap between groups is proportional to the baseline and shrinks as errors decline with age. Whether we call this an interaction is not a fact about the data, it depends on which scale we assume effects add up linearly on. Additive on the log scale means multiplicative (and non-parallel) on the count scale, and vice versa.

This ambiguity extends far beyond this single illustration, as interaction hypotheses are widespread in psychological research and often represent the central target of inquiry (Domingue et al., 2024; McCabe et al., 2022; Rohrer & Arslan, 2021). Whenever researchers ask whether an effect is moderated, or whether it holds only under certain conditions, they are asserting that the impact of one variable depends on another. This count example therefore demonstrates that the reality of such a claim can shift entirely based on the scale on which those differences are computed.

In psychology, this scale dependence has a characteristic source in measurement. Researchers routinely conceptualize constructs such as cognitive ability or symptom severity as continuous latent dimensions (Embretson & Reise, 2000; Wagenmakers et al., 2012). Those dimensions are rarely observed directly; what reaches the data are bounded, discrete quantities, such as counts, ratings, and accuracy scores that carry floors, ceilings, and chance levels. Each measurement scale imposes its own mapping from the latent dimension to the observed response. Where the mapping compresses, a constant effect on the latent dimension appears as a shrinking effect on the observed response: a group or condition closer to a bound has less room to move, and can look less affected by a manipulation, than a group or condition farther away from bounds (e.g., Figure 1, left panel).

Generalized linear models (GLMs) make this mapping explicit and give it a standard vocabulary (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972), separating two choices that are easy to conflate, (1) the outcome family, which describes the response, and (2) the link function, which sets the scale on which effects combine (Box 1).

Box 1: Generalized linear models: outcome family and link function

The *outcome family* is the distribution assumed for the response. It defines which values the outcome can take and how observations vary around their expectation. The *link function*, $g(\cdot)$, maps the expected outcome, $\mu_i = E(Y_i)$, onto the linear predictor, η_i, the scale on which the model is additive. With two predictors and their interaction term,

$$g(\mu_i) = \eta_i, \qquad \eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The inverse link, $g^{-1}(\cdot)$, maps this additive structure back onto the expected response. Its familiar role is to keep expected values within the admissible range of a constrained outcome: $\exp(\eta)$ keeps expected counts positive, and the inverse logit keeps probabilities between 0 and 1. Less obviously, this also fixes what the absence of an interaction, $\beta_3 = 0$, means in the fitted model. The product-term coefficient β_3 is therefore an interaction on the link scale. A response-scale interaction is a different quantity: a difference between expected differences on the observed outcome, for example, whether the group difference in expected errors changes with age in Figure 1. Unless the link is the identity, the two quantities can disagree (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Each family has a canonical link, which is the one that arises directly from the family's mathematical form, but it is a convenient default and other links are legitimate. A Gaussian linear model is the special case that combines a Gaussian family with an identity link, so the additive scale and the response scale coincide and the scale question is invisible. The model behind Figure 1 is a Poisson family with a log link. The family respects that errors are non-negative counts, and the link places additivity on the log scale.

Often the outcome family is chosen with care while the link function is left at its software default, fixing the scale of additivity without deliberation. Two models can share a family and still place additivity on different scales. For example, a Gamma family suits reaction times data;

within that family, a log link makes effects additive in log milliseconds, a proportional change, and the canonical inverse link makes them additive in reciprocal means, a change in speed. A binomial family fits correct-versus-incorrect responses and leaves the link open among logit, probit, chance-corrected, and other forms.

The scale dependence of interactions has long been recognized. Loftus showed that interactions involving response probabilities are meaningful only relative to a mapping between observed and theoretical scales (Loftus, 1978), and Cox formalized interaction as a model-relative concept tied to transformation and interpretation (Cox, 1984). Busemeyer and Jones carried the same point into the measurement model, where the conclusion about how two variables combine depends on the model assumed to link responses to the quantities they measure (Busemeyer & Jones, 1983). Lubinski and Humphreys showed that the specification choices alone can produce the appearance of an interaction where the generating process contains none (Lubinski & Humphreys, 1990).

More recently, Wagenmakers and colleagues separated removable from nonremovable interactions (Wagenmakers et al., 2012). A removable interaction can be undone by a monotonic transformation of the dependent variable, which makes it a property of the measurement scale, whereas a nonremovable one (typically a cross-over) persists under any such transformation. Rohrer and Arslan showed that the same interaction can be positive, null, or negative depending on the scale chosen for the outcome, so the appropriate test follows from the scale on which the hypothesis is posed (Rohrer & Arslan, 2021). Rimpler and colleagues added that a product term assumes one predictor's effect changes at a constant rate across levels of the other, and so misrepresents any dependence that does not take that form, such as a threshold effect (Rimpler et al., 2026). In GLMs specifically, the product-term coefficient refers to the link scale and does not equal the interaction on the response scale, so a question about the response scale should be answered with marginal effects (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026).

Domingue and colleagues demonstrated that outcome-model misspecification can produce false interaction findings for binary, count, and censored outcomes (Domingue et al., 2024), and Liddell and Kruschke showed that treating ordinal responses as metric can inflate error rates for interactions (Liddell & Kruschke, 2018). Czado and Santner, and later Yu and Huang, studied the inferential consequences of link misspecification in binary GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019). Beyond the scale question, interaction estimates are fragile for other reasons, including small effects, low reliability, and limited power (Baranger et al., 2023; Castillo et al., 2026).

This literature spans nearly five decades, but whether applied practice has absorbed the warning is still unaddressed. The present article approaches that gap from two directions. We first document current practice with a preregistered review of recent empirical articles in five prominent psychology journals, quantifying how often interactions are tested, how often the outcome family and link function are made explicit, and which response formats predominate. We then develop a framework that separates outcome family, link function, and target metric, and work it through three common response formats, namely forced-choice accuracy with a chance floor, sum scores, and within-family link choices such as logit versus probit, using simulations to quantify how often an inadequate link turns a purely additive structure into a statistically significant interaction and whether standard diagnostics detect the misspecification.

These simulations take up the case in which the outcome family is already broadly plausible and the link alone sets a disputable no-interaction baseline, the common situation for bounded and chance-limited psychological outcomes. Work on misspecification has largely varied the outcome model as a whole (Domingue et al., 2024; Liddell & Kruschke, 2018), whereas here the family stays fixed and the link carries the entire disagreement about what counts as no interaction. An interaction that appears on a scale other than the generating one is then a genuine mathematical property of that scale (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022), correct where it was computed and troublesome only when the claim is carried to a different scale. Because the generating scale is known in a simulation, we can call such a result a pseudo-interaction, an interaction that is zero on the generating scale and nonzero on another fitted scale because the two encode different additivity baselines. In empirical data, where the generating scale is unknown, we describe interaction conclusions as link-sensitive, conditional on the fitted scale, or unstable across plausible links. Table 4 summarizes these insights into a practical workflow for applied research.

Technical details are reported in Supplement A; a broader precomputed sensitivity atlas varying sample size, main-effect magnitude, the intercept, and selected design parameters is reported in Supplement B.

Current Practice in Psychological Interaction Testing: A Preregistered Review

The review asks how often psychological articles test interactions, and how often they make the outcome family and link function explicit enough for a reader to know which interaction claim was tested. To our knowledge, it is the first review to quantify link-function reporting in psychological interaction tests. The coding is descriptive, so it documents reporting practice rather than the correctness of any substantive conclusion.

Review Method

The protocol was preregistered before any record was screened (https://osf.io/bdu73/), together with the coding data dictionary. The essentials are given here; the full protocol, sampling frame, and reliability are in Supplement A.

Following the preregistration, we sampled one journal from each of five areas: *Psychological Science* (general), *Journal of Experimental Psychology: General* (experimental), *Journal of Personality and Social Psychology* (social), *Developmental Science* (developmental), and *Psychological Medicine* (clinical). The five were selected on preregistered discretionary criteria: a first-quartile 2023 Impact Factor, a broad rather than narrowly specialized scope, not primarily publishing reviews, and strong reputation in their field, as jointly judged by the coauthors. From each journal's full set of 2024 articles in Scopus, records were screened in a fixed random order until 40 eligible interaction-testing articles were reached, for a preregistered total of 200. Because screening stopped once 40 interaction-testing articles were reached, the number of records screened, and therefore the denominator of the eligibility proportion, varied across journals. Eligible articles were empirical reports of original research; reviews, meta-analyses, editorials, theoretical and qualitative-only articles, replication-only studies, and psychometric validation studies were excluded. Detailed coding was applied only to articles testing at least one interaction, and only to observed-outcome analyses: articles whose relevant tests were purely latent-variable or structural-equation models fell outside the coding scheme (12 such exclusions).

Each interaction-testing article was independently coded by two researchers, with disagreements resolved by discussion or by consulting the other coauthors. Generative AI tools were used only as an optional aid to locate passages within an article; all coding used in the review is human.

For each article, the coders recorded five things: whether at least one tested interaction used a non-identity link; whether the link function was explicitly reported; whether at least one interaction applied an incorrect identity-link analysis; whether at least one such interaction was statistically significant, or interpreted analogously in a Bayesian analysis; and which response-variable types were analyzed.

By a formally incorrect identity-link analysis we mean a Gaussian linear regression with an identity link applied to an outcome with formal constraints, such as bounds, discreteness, positivity, or a known chance level. This is a conservative screening flag, applied only when the constraint was identifiable from the report and does not imply the substantive conclusion is wrong. The label is defined relative to the constrained process that generates the outcome (if the observed score were explicitly justified as the target metric, identity-link additivity would instead be a defensible modeling choice, a distinction we develop in the following sections).

The full operational rules the coders followed, including borderline cases, are tabulated in Supplement A. Interrater agreement on the double-coded records was high (Cohen's κ between 0.66 and 0.81); per-variable agreement and coder margins are reported in Supplement A. Preregistered variable definitions, operational coding rules, variable-specific agreement, and by-journal results are also reported in Supplement A, Section S2.

Review Findings

First, interaction testing was common. Screening yielded 331 eligible empirical articles, of which 200 tested at least one interaction, 60.4%. Under the stopping rule this proportion describes the records screened rather than a fixed-denominator sample, so we report it descriptively and treat the per-journal proportions as the more informative summary: these ranged from 38.1% in Psychological Medicine to 85.1% in Journal of Experimental Psychology: General.

Second, model-scale reporting was uncommon. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6], and only 46 explicitly reported the link function, 23%, 95% CI [17.7, 29.3]. Thus, in about 77.0% of interaction-testing articles, the link function was not explicitly stated.

Even the 46 explicit cases rest on a rule looser than the preregistered one, which required a declaration of the link and treated naming a family or a generalized linear model as insufficient. In coding we also accepted link-specific model names, because in practice they identify the link: “logistic regression” names the logit and “probit regression” the probit. We also accepted “Poisson regression”, which names a family whose standard link is the log. The stricter preregistered rule would give a lower count.

Third, constrained outcomes were often analyzed on the identity scale. Among the 200 interaction-testing articles, 144 included at least one formally incorrect identity-link analysis of a constrained observed outcome, 72%, 95% CI [65.4, 77.8]. As noted, this is a conservative screening count, flagged only when the constraint was clear from the report. Of these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]. This is not a comparison, because significance was coded only within the flagged analyses; that is, 62.0% of all interaction-testing articles combined a constrained outcome, an identity-link analysis, and a significant interaction claim. This is the same concern raised by work on outcome-model mismatch in interaction testing (Domingue et al., 2024; Liddell & Kruschke, 2018). Table 1 summarizes these results.

The pattern was present across the sampled journals. In Supplement A, the proportion of

Table 1

Descriptive summary of the preregistered review. Formally incorrect identity-link analyses of constrained observed outcomes are cases where the reported outcome had formal constraints that make identity-link additivity formally incorrect as a generative model unless explicitly justified as an approximation or as the target observed-score metric. The interval on the eligibility row is reported for completeness only: screening stopped at 40 interaction-testing articles per journal, denominator was not fixed by design.

Quantity	n	Denom.	%	95% CI
Eligible empirical articles	331	331	100.0%	[98.9%, 100.0%]
Eligible empirical articles testing at least one interaction	200	331	60.4%	[55.1%, 65.5%]
Interaction-testing articles using at least one non-identity link	58	200	29.0%	[23.2%, 35.6%]
Interaction-testing articles explicitly reporting the link function	46	200	23.0%	[17.7%, 29.3%]
Interaction-testing articles with formally incorrect identity-link analyses of constrained observed outcomes	144	200	72.0%	[65.4%, 77.8%]
Formally incorrect identity-link cases with at least one significant interaction	124	144	86.1%	[79.5%, 90.8%]
Eligible empirical articles not testing interactions	131	331	39.6%	[34.5%, 44.9%]

formally incorrect identity-link analyses of constrained observed outcomes among interaction-testing articles ranged from 40.0% in Psychological Medicine to 95.0% in Psychological Science. In this sample, Psychological Medicine had the lowest screening proportion and the highest non-identity-link proportion (40.0%). This journal, which uses many outcomes for which non-identity links are standard, contributed the fewest formally incorrect identity-link cases.

The outcomes used in interaction analyses were heterogeneous. The most common broad, non-exclusive categories were binary, accuracy, or proportion outcomes (84 articles), sum scores or composites (66 articles), ordinal, Likert, or rating outcomes (36 articles), response times or durations (30 articles), neural or physiological measures (23 articles), difference or distance scores (17 articles), counts or error counts (9 articles), and correlations or associations (8 articles).

For approximately continuous outcomes, additivity on the observed scale can be a defensible target, whereas for bounded and discrete ones it is a stronger assumption, since treating them as metric distorts estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018). Additionally, bounded outcomes dominated the corpus, since the first three categories together appeared in 152 of the 200 articles (76.0%). Therefore, the worked examples focus on binomial and aggregated responses, including sums of binary or ordinal items.

Table 2

Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.

Outcome type	n	Denom.	%
Binary / accuracy / proportions	84	200	42.0%
Sum scores / composites	66	200	33.0%
Ordinal / Likert / ratings	36	200	18.0%
Response times / durations	30	200	15.0%
Counts / error counts	9	200	4.5%
Neural / physiological measures	23	200	11.5%
Correlations / associations	8	200	4.0%
Difference / distance scores	17	200	8.5%

What the Present Review Shows, What It Does Not

As the first review to quantify link-function reporting in psychological interaction tests, it suggests three points. First, interaction testing is pervasive, in every journal sampled and in 60% of the eligible empirical articles screened. Second, the scale of additivity is usually left implicit: only 23% named the link function, and even that count leans on counting model names (such as “logistic regression”) rather than the link function per se. Third, constrained outcomes are routinely analyzed on the identity scale: 72% of interaction-testing articles applied an identity-link analysis to an outcome with formal constraints, and most of these reported a significant interaction. In leading journals, then, interaction tests are common while the scale on which no interaction is assumed is rarely stated.

Nevertheless, a descriptive review cannot establish how often published interactions are scale artifacts, because that judgment requires knowing the scale on which the data-generating process is additive. The remainder of the paper therefore turns to settings where the generating process is known by construction (i.e., simulations).

The Link Function Problem in Three Common Cases

To demonstrate the consequences of link misspecification we conduct three simulation studies. The three simulations ask whether a purely additive data-generating process on one scale is detected as a statistically significant interaction when modeled on another scale:

1. *Forced-Choice Accuracy*: This case holds the binomial outcome family fixed and evaluates whether the link function must explicitly encode a task-implied chance floor (e.g., a .50 guessing threshold).
2. *Within-Family Link Choice (Logit vs. Probit)*: This case holds both the binary family and the 0-to-1 bounds fixed, testing whether choosing between these two standard link functions alters the results.
3. *Sum Scores*: This case holds the latent generating process fixed and evaluates whether the bounded total shows an interaction that the latent scale does not.

The simulation scenarios were chosen to represent plausible psychological data patterns. Each reproduces a common interaction-testing situation: two main effects that are well established and pretty large, for example a group difference combined with a condition effect or a developmental trend, and the question of whether one effect moderates the other. The two main effects were always tuned strong enough to push expected values close to a floor, ceiling, or chance level while keeping them clearly inside the observed range. This is the region where link curvature acts most strongly.

Each simulation applies the same design. Data are generated with no product term on the generating scale, and models assuming additivity on that scale and on plausible alternative scales are fitted to the same data. Every fitted model yields a product-term rejection rate. We called *nominal rejection rate* the rejection under a matched generating-scale model, whereas rejection under a mismatched-scale model is the *pseudo-interaction detection rate*. In the next section, a final simulation examines whether standard diagnostics reliably identify the misspecified link.

Forced-Choice Accuracy and the Non-Zero Chance Floor

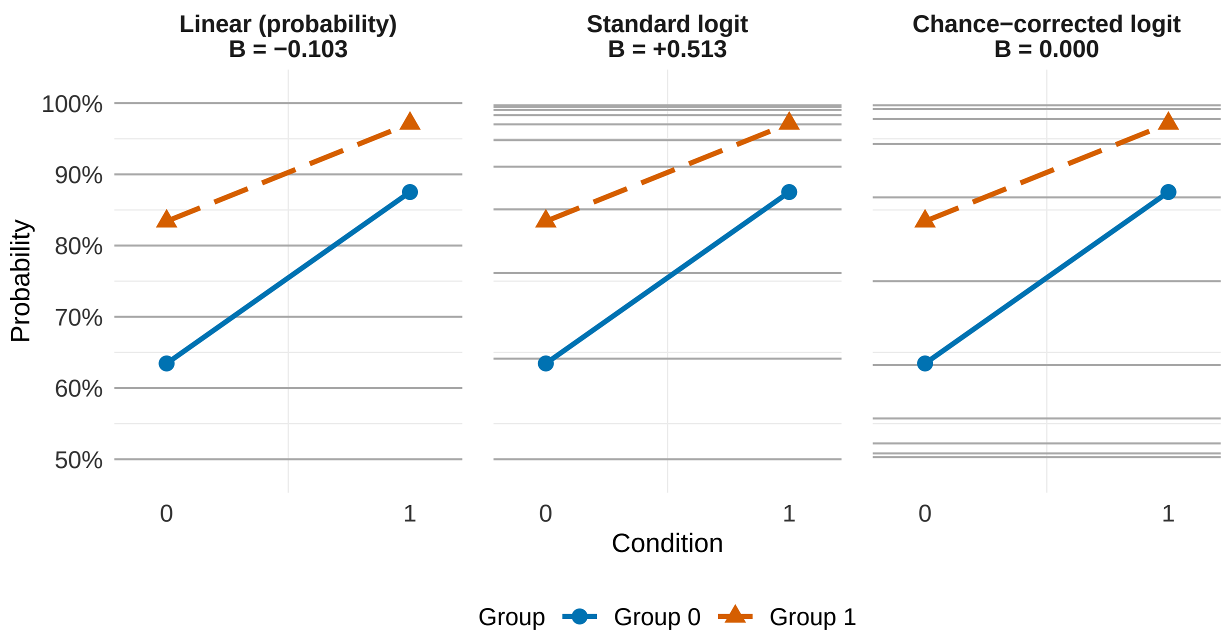
If a participant answers Y_i of k trials correctly, a binomial sampling model, $Y_i \sim \text{Binomial}(k, p_i)$, is the natural starting point. The family requires the expected accuracy to lie between 0 and 1. When guessing is a plausible process for the task, the range that matters is narrower, since a participant who knows nothing still gets a share c of the trials right, $c = .50$ in a two-alternative or true/false task and $c = 1/m$ with m alternatives, so expected accuracy lies between c and 1, even though observed proportions can fall below c through sampling variation. Accuracy data of this kind are analyzed in several ways in practice, on the raw proportion scale with a Gaussian identity model, with a standard binomial logit or probit, and, less often, with a link that encodes the chance floor (Figure 2).

These specifications differ in which of the two ranges they impose. The Gaussian identity model imposes neither, since it ignores the trial process, assumes additivity in percentage-correct units, and can predict outside the 0-to-1 range. The other two keep the binomial family and differ only in the link. The standard logit or probit, $p_i = F(\eta_i)$ with F the logistic or standard-normal CDF, holds expected accuracy between 0 and 1 and places the lower asymptote at 0, whereas the chance-corrected link maps the linear predictor onto the narrower interval from c to 1, $p_i = c + (1 - c)F(\eta_i)$, which for $c = .50$ becomes $p_i = .50 + .50\,F(\eta_i)$.

An interaction term in the standard model therefore tests additivity over the full 0-to-1 range, whereas the same term in the chance-corrected model tests additivity on a performance-above-chance scale.

Figure 2

A minimal 2×2 *design crosses a condition (x-axis) with a group (color), shown on three scales. The four cell probabilities are identical in every panel; only the scale on which the interaction is evaluated changes. The data were generated with no condition-by-group interaction on the chance-corrected logit scale, which has a* .50 *chance floor, so the interaction coefficient is exactly zero on that scale. The same cells imply a positive interaction on the standard logit scale and a negative interaction on the raw probability scale. Light horizontal lines are equally spaced on each panel's own link scale, so equal steps on that scale map to unequal steps on the probability axis.*



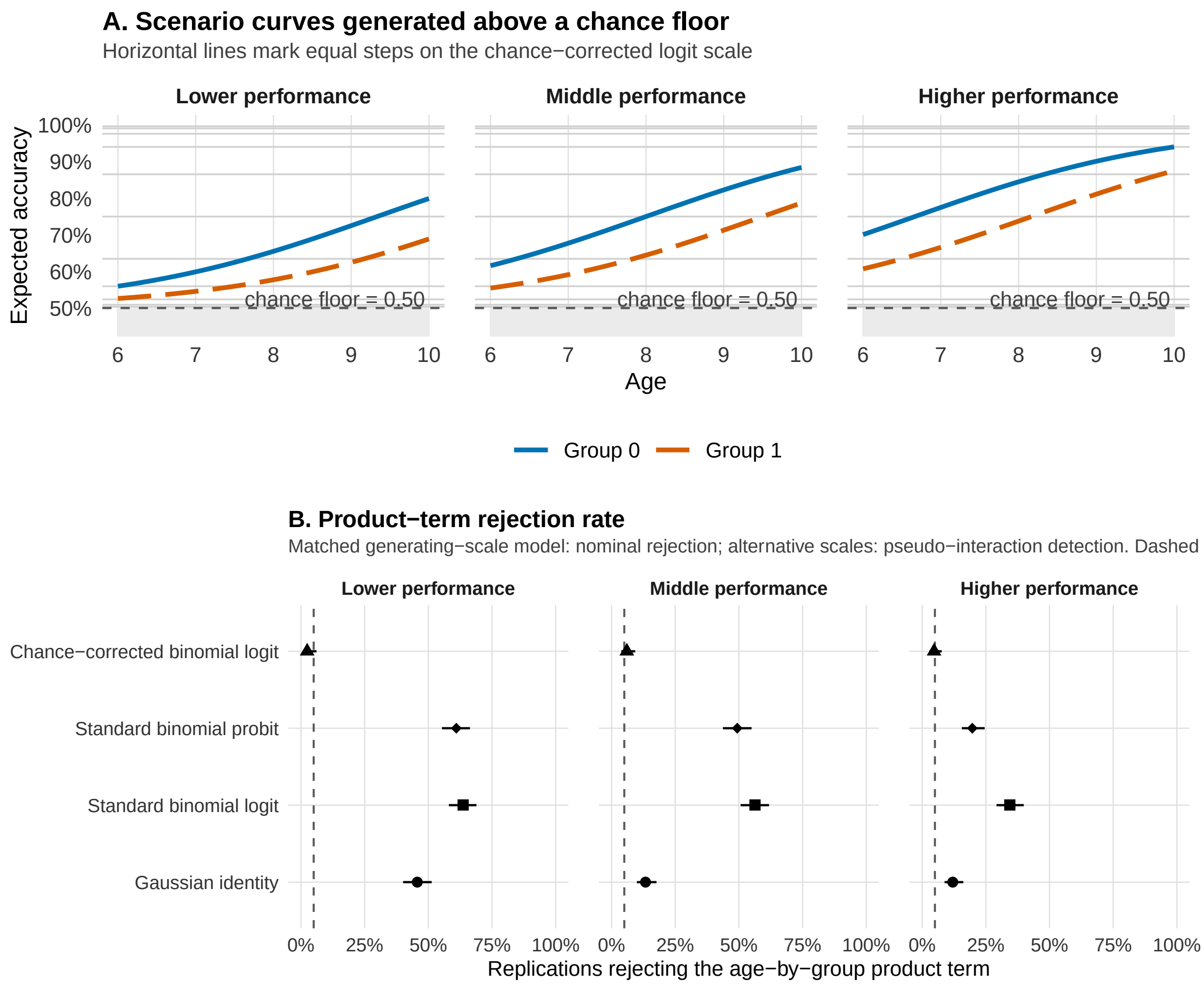
The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. In the simplest case, the chance level can be fixed by design, for example $c = 0.50$ in a two-alternative task. In other cases, analysts may estimate the lower asymptote or report sensitivity analyses over plausible values of c. Lapses, response biases, item heterogeneity, changes across trials, or systematic below-chance performance may instead require richer psychometric solutions such as item-response models (e.g., Embretson & Reise, 2000; Knoblauch & Maloney, 2012; Wichmann & Hill, 2001). For example, the 3-parameter logistic Item Response Theory model includes an item-specific guessing parameter rather than a single chance level fixed at the design level (Embretson & Reise, 2000). The chance-corrected link should thus be seen as a design-level approximation, not a substitute for item-level measurement models when the data and scientific question call for that additional structure.

Simulation 1: Design and Results

In each of 300 replications we generated accuracy for $N = 250$ participants with $k = 20$ trials each, from a chance-corrected logit model with a .50 floor, a positive age effect, a group difference, and no age-by-group product term on the generating scale. Three scenarios differ only in overall performance, set by shifting the intercept: in the *lower-performance* scenario the two groups lie close to the .50 floor (expected accuracies roughly .53 to .80 across ages 6 to 10), in the *middle-performance* scenario they occupy the middle of the above-chance range (roughly .55 to .88), and in the *higher-performance* scenario they approach the ceiling (roughly .61 to .94). Each simulated dataset was analyzed with four models: a Gaussian linear model on the observed proportions, standard binomial-logit and binomial-probit GLMs on the correct-response counts, and the chance-corrected binomial-logit GLM that matches the generating scale. We recorded how often each product-term test rejected at $\alpha = .05$. The matching model supplies the nominal rejection rate, while the three alternative scales supply pseudo-interaction detection rates. Figure 3 shows the simulation results; the per-scenario rates and the convergence counts are tabulated in Supplement A.

Figure 3

Simulation 1: forced-choice accuracy with a chance floor. Data are generated with no age-by-group product term on the chance-corrected logit scale. Panel A shows the resulting accuracy curves across three performance levels; the grey region lies below the .50 chance floor. Panel B shows product-term rejection rates by fitted model and scenario. The matched chance-corrected model remains near nominal alpha, whereas the alternative scales produce pseudo-interactions. The dashed line marks alpha = .05.



The chance-corrected model rejects between 2% and 6% of the time across the three scenarios. Pseudo-interaction detection is highest where performance lies near the floor: in the lower-performance scenario the standard logit rejects 64% of the time, the standard probit 61%, and the Gaussian identity 46%. Detection falls as performance moves away from the floor, reaching 34% for the logit, 20% for the probit, and 12% for the Gaussian identity in the higher-performance scenario. The same qualitative sensitivity to the chance floor also appears at chance levels 0.25 and 0.33 (Supplement B).

Although the chance-corrected model rejects in 2% of the time in the lower-performance scenario, the product term there is weakly identified. As expected accuracy approaches chance, the curve that maps the above-chance scale onto observed accuracy flattens, so widely different values on that scale imply almost the same expected accuracies, and the data cannot distinguish them. Indeed, in the lower-performance scenario only 164 of the 300 replications produced a usable test of the product term. The rejection rate is therefore computed on those replications, and since failures concentrate where the data are least informative about above-chance differences, they are a selected subset. Restricting all four models to those 164 replications puts them on identical datasets and leaves the pattern unchanged. The chance-corrected model rejects in 2% of them, against 45% for the Gaussian identity, 63% for the logit, and 62% for the probit.

Within-Family Link Choice

The forced-choice case involved a link that was plainly wrong for the task, because it placed the lower asymptote at 0 when the design fixed it at chance. But the link-function problem can also arise in less obvious cases. Logit and probit are both standard for binary data, respect the same 0-to-1 bounds, and usually produce very similar fitted probabilities. Their choice is therefore often guided by convention, ease of interpretation, or implementation, with little expected impact on substantive conclusions (Agresti, 2002; Yu & Huang, 2019). This makes them a special case for illustrating the problem, as even two virtually interchangeable links define slightly different scales of additivity. Equal differences under one link are not exactly equal under the other, especially toward the tails, and this alone can alter the estimated product term.

The logit parameterization also gives coefficients a direct log-odds interpretation, whereas probit coefficients are expressed on a standard-normal latent scale, making it a choice of election when modeling psychological quantities such as ability, signal strength, and response tendency

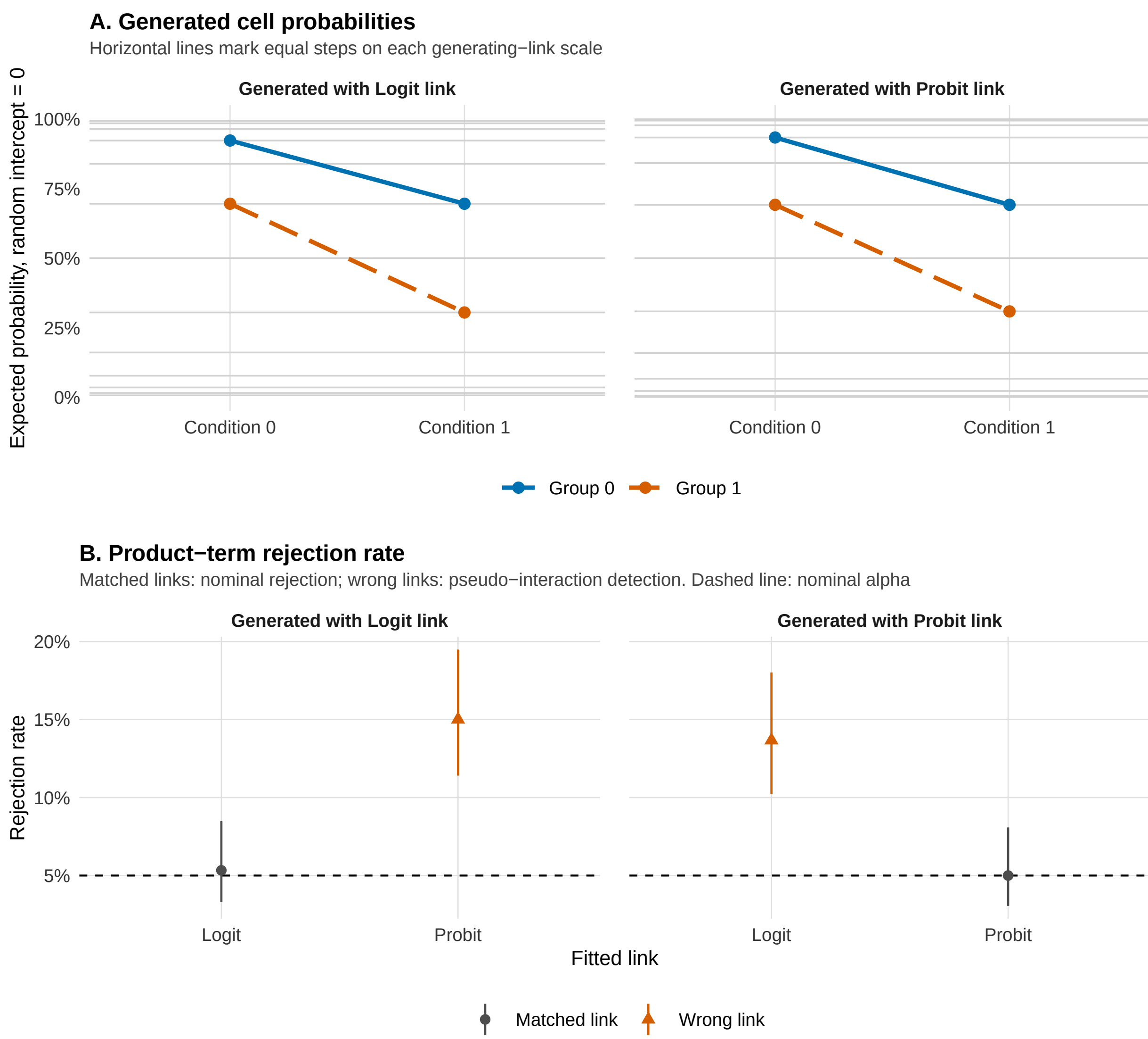
(Dunn & Smyth, 2018).

Simulation 2: Design and Results

In each of 300 replications we generated repeated binary responses for $N = 700$ participants in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept (latent intraclass correlation 0.30). The quite large sample was chosen to make the relatively small systematic discrepancy between logit and probit links detectable, rather than to represent a typical psychological study. The condition-by-group interaction term was zero on the generating link scale. We crossed the generating link with the fitted link, using logit and probit in both roles, and fitted every model as a random-intercept GLMM, recording how often each returned a significant condition-by-group interaction term at $\alpha = .05$. Figure 4 shows the results; the full crossed-link rates are tabulated in Supplement A.

Figure 4

Simulation 2: logit versus probit in repeated-trial random-intercept GLMMs. Data are generated with no condition-by-group product term on the generating link scale. Panel A shows the corresponding cell probabilities under logit and probit links. Panel B shows product-term rejection rates for each generating–fitted link combination. Matched links remain near nominal alpha, whereas wrong links produce pseudo-interactions. Error bars are Wilson 95% confidence intervals; the dashed line marks alpha = .05.



When the fitted link matches the generating link, the nominal rejection rate stays near .05:

5% for logit fitted to logit-generated data and 5% for probit fitted to probit-generated data. Crossing the links raises the pseudo-interaction detection rate to 14% when a logit model is fitted to probit-generated data and 15% when a probit model is fitted to logit-generated data, roughly three times the nominal rate. The wrong-link detection rates are smaller than in the forced-choice case, and they appear among links that share the same family, the same 0-to-1 bounds, and nearly overlapping fitted curves in the middle of the probability range, the setting many applied analysts treat as interchangeable. Under the wrong link the same cell probabilities imply a small but deterministic nonzero product term on the fitted scale. Supplement B varies the intercept, which shifts where the four cell probabilities sit between 0 and 1, and shows that detectability changes non-monotonically with it, so the same link mismatch is nearly undetectable at some probability levels and much easier to detect at others.

A result that is stable across both links is easier to defend; one that appears only under a single link should be reported as link-conditional. Supplement A, Section S5, reports how large that fixed product term is. Taking the cell probabilities implied by each generating model and re-expressing them on the other link scale gives a product term of -0.112 for logit-generated data fitted with a probit and 0.216 for probit-generated data fitted with a logit, against exactly zero whenever the links match. The rejection rates above are therefore the power to detect these effects.

Sum Scores

Many psychological outcomes are computed as total sum scores across questionnaire items, task items, symptoms, errors, or ratings. In the two cases above the outcome was a directly observed response, and only the link was in question. A sum score is different, because the score itself is already the result of an analytic choice. What is observed are the individual item responses, which are binary or ordinal; the total is a summary of them, produced by a scoring rule that gives every item the same weight and every point of the scale the same width. In psychological research the construct of interest is often the underlying, generally continuous dimension, so the total can be seen as a remapped version of that continuum. The link question then arrives together with a prior question about which metric the claim is about.

The bounds are crucial here, because the construct may have no floor or ceiling while the observe total score has both. The mapping is best read as an exchange rate between the two scales, that is, how much of the latent dimension is needed to gain one raw-score point. That rate is not constant. Near the middle of the scale a small latent shift is enough to move the total by a point, while near either bound a wide stretch of the latent dimension is compressed into the last available point, because few items remain that the respondent can still change. To illustrate, with nine items scored 0-3, moving from 13 to 14 points requires less change on the latent dimension than moving from 26 to 27. A constant latent effect therefore produces smaller raw-score differences as the total approaches an end, and the same reasoning applies wherever the instrument is unevenly informative along the latent dimension, so compression is not confined to the endpoints.

Bounded, discrete, and skewed distributions are common in psychological data (Micceri, 1989), and the consequences of analysing ordinal responses as if they were metric have been documented in detail (Liddell & Kruschke, 2018). Fitting a Gaussian identity model to a total assumes instead that a predictor moves the score by the same number of points at every level of the other predictor. A group/condition sitting near a bound moves by fewer points even when the data-generating process on the latent dimension is the same in both groups/conditions. Behaviour-genetic research has met the same problem when testing whether genetic effects vary across environments, since those interactions are usually tested on summed questionnaire scores: analysed at the item level, the same data show that the total can conceal real interactions or present spurious ones through properties of the observed score (Molenaar & Dolan, 2014). This is the sum-score form of outcome-model misspecification and target-metric mismatch (Domingue et al., 2024).

The problem we raise sits next to a recent, wider debate about whether sum scores should be replaced by factor scores or latent-variable models. Some argue that sum scores are transparent, reliable enough for many purposes, comparable across studies, and tied to how instruments are scored in applied settings (Widaman & Revelle, 2023); the counterargument is that a total encodes strong and usually untested assumptions about how items represent the construct (McNeish, 2023;

McNeish & Wolf, 2020). While we do not enter the debate, we add a narrower point that was left aside: a model of the manifest total defines the absence of an interaction in raw-score points, so an interaction found there is an interaction on the score, and it becomes a claim about the construct only under the additional assumption that the two are linearly related.

When item-level data are available and the claim concerns latent constructs, an explicit measurement model such as SEM, item-level ordinal regression, or IRT-inspired specifications could be preferred, since it addresses the mapping directly, though at the price of moving the assumptions into the measurement model (Bürkner & Vuorre, 2019). Often, however, researchers deliberately work with the total score for the practical reasons listed above. An option is to set a family that respects the bounds and the discreteness, for example modelling the total as a bounded count with a beta-binomial likelihood (Wilcox, 1981). However, we suggest a deliberately unusual option that moves only the link: keep the observed total score under a Gaussian likelihood, but rescale the score to the proportion of the maximum and use a probit link function, as in Simulation 3 below. This is an *ad hoc* device, not a standard practice, but it illustrates whether the interaction conclusion changes if additivity is placed on a nonlinear scale that compresses the expected values toward the score bounds. It remains a model of the total, it will not recover the construct, and the Gaussian likelihood is still misspecified in its variance, since a bounded score has a variance that depends on its mean: what the probit link corrects is only the scale on which additivity is defined, not the distributional form. Read against the sum-score debate, it is a precaution for researchers who have decided to analyse totals anyway, but acknowledge that the interaction still needs a scale on which additivity is plausible.

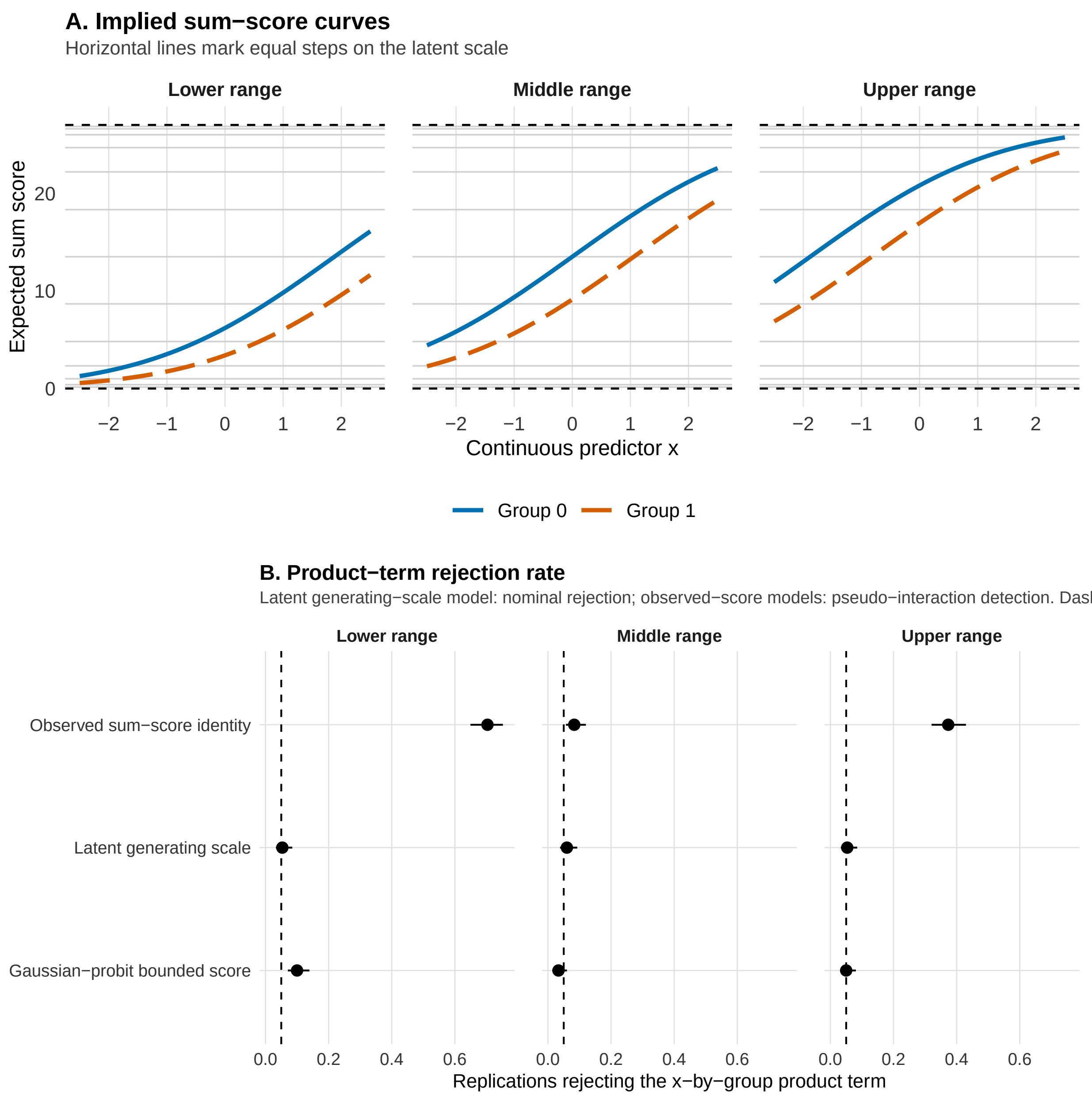
Simulation 3: Design and Results

The bounded-score probit model was compared with two natural alternatives: the observed total and the latent scale it approximates. In each of 300 replications, we generated data for $N = 600$ participants from a continuous latent variable with a continuous predictor x, a group effect, and no x-by-group interaction. The latent values generated 9 ordinal items scored 0 to 3, which were summed to a total ranging from 0 to 27. Item thresholds were shifted across three scenarios so that expected totals fell in the middle, lower, or upper part of the score range.

We analyzed each total in three ways: a Gaussian identity model of the observed sum score, a Gaussian-probit bounded-score model, and the true latent generating scale. The bounded-score model applies a probit link to a continuity-corrected score proportion and rescales predictions to the original sum-score metric. We used probit because its standard-normal linear predictor provides a simple correspondence with the latent scale. This choice is still substantive, as shown in the previous section. The latent model serves as an idealized benchmark available only in simulation. We recorded the proportion of significant x-by-group interactions at $\alpha = .05$. Results are shown in Figure 5, with per-scenario rates in Supplement A, Section S4.

Figure 5

Simulation 3: sum scores as bounded and discrete outcomes. Data are generated with no x-by-group product term on the latent scale. Panel A shows how the same additive latent structure maps onto sum scores in the middle, lower, and upper parts of the score range. Panel B shows product-term rejection rates for the latent benchmark and two observed-score models. The latent model remains near nominal alpha, whereas the observed-score models can produce pseudo-interactions. The dashed line marks alpha = .05. The latent benchmark is available only in simulation.



The latent benchmark remained close to the nominal .05 rejection rate in every scenario, from 5% to 6%. The identity model behaved similarly only in the middle of the score range, where the sum score was approximately linear in the latent variable (8%). Rejection increased as expected totals moved toward a bound, reaching 37% in the upper range and 70% in the lower range. Compression of the latent scale makes the same latent group difference appear to change across levels of the predictor. The bounded-score probit model remained close to nominal alpha across scenarios, from 3% to 10%. It respects the score bounds, although it is still a model of the manifest total.

This pattern mainly depends on where observations fall within the score range. Across the broader scenarios in Supplement B, the position of expected scores within the range affected identity-model rejection much more than sample size, item count, or latent dispersion. Raw-score additivity can approximate latent-scale additivity when scores remain away from the bounds. Compression near a bound can instead create an apparent interaction from a constant latent effect. The bounded-score probit model can therefore serve as a useful sensitivity analysis. For claims about interactions between latent constructs, an explicit measurement model provides a more direct approach, for example SEM, item-level ordinal models, or IRT-inspired models. These approaches still depend on assumptions about thresholds, links, loadings, and latent-response scales, but they make those assumptions part of the interaction model.

Can Standard Diagnostics Catch a Wrong Link?

The simulations above assume the analyst does not question the fitted model. In practice, a researcher who finds an interaction may run model checks before reporting it. We examine the three checks an applied psychologist is most likely to run, each representing a different strategy. A Pregibon-style added-term test probes the assumed link, residual diagnostics screen for misfit of any kind, and AIC ranks a set of candidate models. Estimating the shape of the link from the data through a parametric family of links is a more direct strategy, and it is rarely used in psychology (Aranda-Ordaz, 1981; Czado & Santner, 1992; Stukel, 1988). None of them identifies the scale on which the no-interaction claim is defined. At best they point to the link that fits the data best among the alternatives supplied.

A link test of the kind proposed by Pregibon targets link adequacy directly, in the added-variable form used here (Pregibon, 1980). The model is fitted, the fitted linear predictor $\hat{\eta}$ is extracted, and the model is refitted with $\hat{\eta}^2$ added as an extra predictor. If the link is adequate the squared term carries no additional information and should be non-significant; a significant term signals curvature that the current link leaves unexplained. The signal is not specific to the link, as an omitted nonlinear term, an omitted covariate, or a missing interaction produce it as well. A non-significant result means only that this diagnostic did not detect a departure in this sample.

Residual diagnostics start from the position of each observed value within the distribution the model predicts for that observation. That position is a number between 0 and 1, and under an adequate model those positions are uniform whatever the family, which gives residual plots a common reference distribution and makes misfit visible as departure from uniformity, as over- or underdispersion, or as residual structure over fitted values and predictors. Dunn and Smyth pass the position through the normal quantile function, so their quantile residuals are standard normal under an adequate model (Dunn & Smyth, 1996), while DHARMa obtains the same positions by simulating datasets from the fitted model and leaves them on the 0-to-1 scale (Hartig, 2024). We use the DHARMa implementation in R. As with the link test, several causes can produce departure from uniformity, including link misspecification, wrong family, missing nonlinearity, missing interaction, and target-metric mismatch.

Information criteria compare a supplied set of models. We use AIC throughout, and because our candidate models keeps the response, family, and structural formula fixed and varies only the link, the ranking depends entirely on the log-likelihood. Importantly, the result is restricted to the candidate links that were fitted, and the interaction term is already present in each model and can absorb part of the curvature produced by the wrong link. A misspecified model may therefore fit almost as well as the target model while still placing additivity on the wrong scale.

Diagnostic Simulation

The five scenarios reuse examples already introduced in the paper (Table 3). Each keeps the outcome family and the interaction formula fixed and changes only the link, so the fitted model is wrong about the scale on which effects add and about nothing else. *Count errors* reproduces the Poisson-log example of Figure 1 and fits it with an identity link. *Chance-floor accuracy* takes the lower-performance scenario of Simulation 1, generated on the chance-corrected logit scale and fitted with a standard binomial logit. *Binary repeated trials* reproduces the probit-generated, logit-fitted case of Simulation 2. *Binary continuous predictor* repeats that case with the two-level condition replaced by a continuous predictor, which allows the residual checks that need variation along a predictor. *Gamma mean response time* generates response times from a Gamma model with a log link and fits the canonical inverse link. The generating parameters for all five are reported in Supplement A.

In each replication, diagnostics were applied to the misspecified interaction model, because this is the model an analyst would usually inspect after obtaining a significant interaction. All rates are computed over the replications in which that model converged, which is every one of the 300 replications except in the count scenario, where the Poisson-identity model returned a usable fit in 281 of them. The question is therefore whether the same model that produces the pseudo-interaction also shows enough evidence of misspecification to warn the analyst.

Each replication was screened with four DHARMa checks, covering residual uniformity, dispersion, residual quantile patterns over fitted values, and residual structure over the focal predictor (Dunn & Smyth, 1996; Hartig, 2024). The last of these takes two forms, a quantile check when the focal predictor is continuous and a comparison across design cells in the categorical 2×2 case. Alongside them we ran the Pregibon-style added-term link check (Pregibon, 1980). Every procedure was applied a second time to the correctly specified interaction model fitted to the same data, which calibrates the flagging rates against a model that is right.

The AIC comparison in each scenario is between the two models already described, the target and the misspecified one. We report the proportion of replications in which AIC favored the target model.

Table 3

Diagnostic simulation across five link-misspecification scenarios. Pseudo-interaction reports how often the misspecified model detected an interaction that was absent on the generating link scale. DHARMa gives the range across the four prespecified residual checks. Pregibon reports the added-term link check. AIC favors target reports the proportion of replications in which the lower AIC favored the target interaction model over the misspecified interaction model. CI = Wilson 95% confidence interval. The Pregibon check is not identifiable in the saturated 2×2 design, where the squared linear predictor is exactly collinear with the existing fixed effects.

Case	Target model	Misspecified model	Pseudo-interaction rate	DHARMa	Pregibon	AIC favors target
Count errors	Poisson log	Poisson identity	0.957 [0.927, 0.975]	0.000-0.883 (4)	0.986	0.986
Chance-floor accuracy	chance-corrected binomial logit	standard binomial logit	0.587 [0.530, 0.641]	0.000-0.060 (4)	0.317	0.813
Binary repeated trials	binomial probit GLMM	binomial logit GLMM	0.127 [0.094, 0.169]	0.000-0.013 (4)	N.A.	0.950
Binary continuous predictor	binomial probit GLMM	binomial logit GLMM	0.073 [0.049, 0.109]	0.000-0.003 (4)	0.043	0.937
Gamma mean response time	Gamma log	Gamma inverse	0.297 [0.248, 0.351]	0.000-0.063 (4)	0.340	0.793

Table 3 shows that the three checks behave very differently. The Pregibon test flagged the count case in 99% of replications, but reached only 34% in the Gamma case and 32% at the chance floor, fell to 4% for the binary case with a continuous predictor, and could not be computed at all in the saturated 2×2 design. The DHARMa checks were weaker, reaching 88% in the count case and at most 6% in the other four, so residual screening responded only where the wrong link visibly distorted the fitted values. AIC favored the target link in every scenario, from 79% in the Gamma case to 99% in the count case, and was the only check that responded throughout.

The two logit-probit cases were the hardest to diagnose. They produced the fewest pseudo-interactions, 13% in the saturated design and 7% with a continuous predictor, and almost nothing detected them, leaving AIC at 95% and 94% as the only informative signal.

The chance-floor scenario is worth a closer look, because it combines a high pseudo-interaction rate with residual checks that are almost never significant, which leaves AIC as the only diagnostic that detects the misspecification. There AIC favored the chance-corrected model in 81% of replications, but the margin is often narrow, and Supplement A reports the full distribution of the AIC difference for every scenario.

Overall, the diagnostics were informative in some settings and weak in others, even when the pseudo-interaction significance rate is pretty high. Under the correct link, DHARMa flagged a problem in at most 4.7% of replications. Pregibon flagged at most 5.3% in the count, chance-floor, and Gamma scenarios, and 4.3% in the continuous-predictor GLMM.

Full results are reported in Supplement A, and Supplement B shows how pseudo-interaction rates, Pregibon detection, and AIC preference change across sample size and the intercept.

From Link Choice to Sensitivity Analysis

The practical implication of this framework is that an interaction analysis should make three choices explicit: the link that defines additivity, the metric the claim refers to, and the sensitivity checks used to assess that choice (Table 4). The observed metric may sometimes be the relevant target. For example, a two-point difference on a clinical or educational score may be the quantity that matters in practice, and additivity in raw-score units may then be reasonable. The key point is that this should be a substantive choice, not the by-product of using a default model. When possible, the link should be justified from theory, task structure, or measurement (Baranger et al., 2023; Castillo et al., 2026).

When link choice can affect the interaction conclusion, we suggest that the main robustness check is to performa a sensitivity analysis across plausible links. Researchers should therefore compare the same target response-scale contrast across alternatives and report whether the result is stable or link-conditional. If it changes, that dependence is itself part of the result (McCabe et al., 2022; Mize, 2019; Mustillo et al., 2018; Rohrer & Arel-Bundock, 2026).

Table 4

Practical starting points for link-aware interaction testing across common psychological outcomes.

Outcome type	Plausible starting point	Minimal sensitivity check
Forced-choice accuracy	Binomial logit/probit; chance-corrected link when the task floor is part of the estimand.	Compare standard logit/probit and chance-corrected links; report convergence and response-scale contrasts.
Binary / proportion outcome	Binomial logit/probit; consider cloglog for asymmetric processes.	Compare logit/probit; inspect calibration and contrasts near bounds or rare-event regions.
Likert / ordinal item	Ordinal logit/probit; metric model only with a clear observed-scale target.	Compare ordinal and metric summaries; report category probabilities or response-scale contrasts.
Sum score / composite	Gaussian identity when many levels stay far from bounds; otherwise consider item-level, ordinal, bounded-count, or bounded-score models.	Inspect bounds, skew, and score concentration; compare with item-level or bounded-score sensitivity models when feasible.
Counts / error counts	Poisson or negative binomial, usually with log link; identity only with an observed-count target.	Check overdispersion and zero inflation; compare count contrasts across plausible count models.
Response time / skewed positive outcome	Lognormal, Gamma log/inverse, Inverse Gaussian, or transformed Gaussian, matched to absolute, proportional, reciprocal, or transformed change.	Compare raw-, log-, and response-scale summaries; inspect tail fit and predicted values.

Discussion

The same data can support different interaction conclusions under different plausible specifications, which is the familiar scale dependence of interactions (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012). In a GLM that dependence is carried by two choices, the outcome family and the link function. Our review found interaction tests to be common in all five journals sampled, the link function usually left implicit, and identity-link analyses applied to many outcomes for which observed-scale additivity is not self-evident.

The simulations then showed what that can cost. With two main effects present, a link that misplaces additivity turned a purely additive generating process into a significant product term even when the outcome family was correct. In forced-choice accuracy the binomial family is correct and the error lies entirely in a canonical link that places the lower asymptote at 0 rather than at the chance level. The logit-probit comparison showed that the problem survives when the response bounds are correct as well, between two links that look alike over most of the range. In the sum-score case the choice extends beyond the link to the outcome itself. A total is a scoring rule applied to item responses, so an interaction estimated on it refers to raw-score points, and adopting a family that respects the bounds and the discreteness of the total still leaves open whether additivity in points stands for additivity in the construct.

This difficulty is of a different kind from the one documented in work on the precision of interaction estimates (Baranger et al., 2023; Castillo et al., 2026). There the obstacle is noise, and more data helps. Here the product term induced by the wrong link is a fixed property of that scale, so a larger sample estimates it more precisely and reports it as significant more often.

The diagnostic simulation showed that ordinary post-fit checks catch this unevenly. Residual checks, Pregibon-style tests, and AIC each detected the misspecification in some scenarios and not in others, and their sensitivity depended on which candidate models were supplied. The chance-floor scenario illustrates this most clearly, since a wrong link produced pseudo-interactions in most replications while the residual checks were almost never significant, and the only check that detected the misspecification was the explicit comparison between the two candidate links, which preferred the correct link by a narrow margin. These diagnostics therefore answer a question about fit within a supplied candidate set, which is a different question from whether an interaction claim holds on the scale the researcher means (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

Three limitations bound the scope of these conclusions. The review was descriptive and covered five reference journals across major areas of psychology in a single year, so its estimates describe reporting practice in high-standard outlets rather than the field as a whole, and a broader mapping across journals, years, and subfields would be needed to support reporting guidance. The review also coded observable reporting features and did not judge, article by article, whether a given interaction was scale-dependent enough to be removable. The simulations, finally, are demonstrations of a mechanism. They target a single regime, in which two established main effects push expected values into the region where link curvature is strongest, and they say nothing about how often that regime arises or about the further complications of real data.

To conclude, interaction claims rest on a scale that is rarely stated. Naming it separates three questions that a software default otherwise settles at once, which family suits the response, which link defines additivity, and which metric the claim is about. The robustness question then takes a definite form, whether the conclusion survives the links that remain defensible. A result that holds across them is robust to that choice. One that does not is still a result, and it should be reported as conditional on the scale on which it was estimated.

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